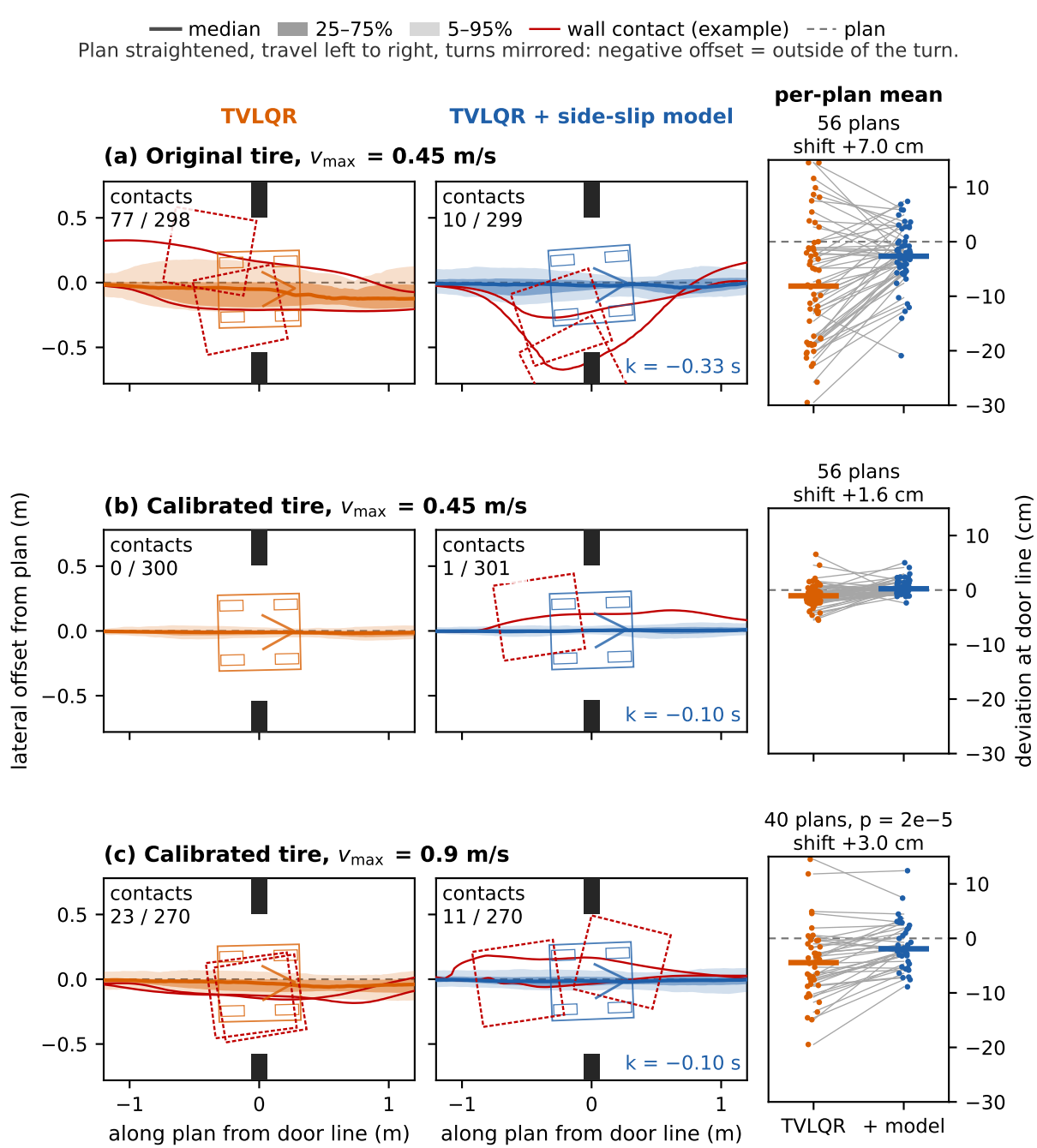

median
25–75%
5–95%
wall contact (example)
plan
Plan straightened, travel left to right, turns mirrored: negative offset = outside of the turn.
TVLQR
TVLQR + side-slip model
per-plan mean
(a) Original tire, v_{max} = 0.45 m/s
contacts
77 / 298
contacts
10 / 299
k = −0.33 s
56 plans
shift +7.0 cm
(b) Calibrated tire, v_{max} = 0.45 m/s
contacts
0 / 300
contacts
1 / 301
k = −0.10 s
56 plans
shift +1.6 cm
(c) Calibrated tire, v_{max} = 0.9 m/s
contacts
23 / 270
contacts
11 / 270
k = −0.10 s
40 plans, p = 2e−5
shift +3.0 cm
lateral offset from plan (m)
deviation at door line (cm)
along plan from door line (m)
along plan from door line (m)
TVLQR
+ model